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NETWORKING

MID EXAM REVISION

Control the Path. Control the Outcome.

Control the **Path.** Control the **Outcome.**

This practical Revision is designed to **evaluate your ability to configure and apply advanced routing concepts** covered throughout the course. It focuses on hands on implementation across multiple routing technologies, including **OSPF**, **EIGRP**, and **BGP**, as well as path control and network optimization techniques.

You are required to configure the network based on the provided topology and complete all configuration tasks using best practices. The exam emphasizes accuracy, and proper verification of routing behavior.

This Revision should be treated as a **supplemental practice exercise** It is not official but rather a structured opportunity to reinforce your understanding and improve your technical confidence before assessments.

Instructions:

- Follow the topology and requirements carefully.
- Apply proper configuration practices and ensure all protocols operate correctly.
- Verify your work using appropriate show and testing commands.

Disclaimer:

This revision is for **training and revision purposes only**. However, completing this revision thoroughly will reinforce every major concept required for success and significantly increase your readiness for the mid assessment.

Network Topology: Download Topology Diagram on Git Hub

Scenario

You are a network engineer responsible for designing and configuring the routing infrastructure of a growing enterprise network.

The organization consists of a central core network, an external service provider domain, and a remote branch office. The network uses multiple routing protocols to reflect real-world enterprise and service provider environments.

The internal enterprise network operates using OSPF with a hierarchical multi-area design to ensure scalability and efficient routing. The enterprise connects to an external autonomous system through dual-homed BGP links for redundancy and traffic engineering.

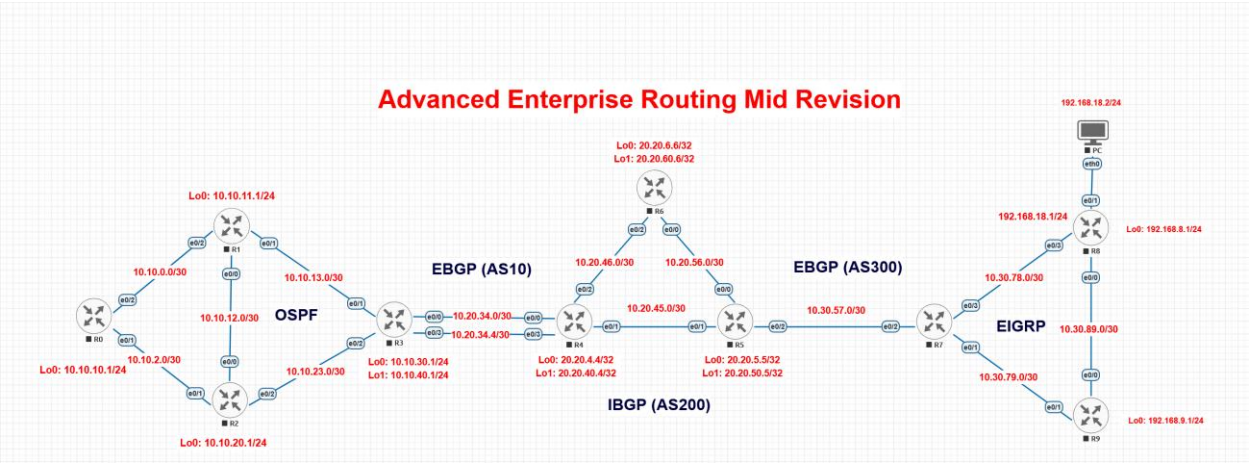
Within the service provider domain, iBGP is used to distribute routing information between core routers. Proper BGP configuration and policy control are required to ensure optimal path selection for both inbound and outbound traffic.

The remote branch network operates using EIGRP and is designed following a hierarchical model where access routers should not act as transit devices. Scalability and fast convergence are critical, and appropriate EIGRP features must be implemented to control query propagation and optimize routing behavior.

Your task is to configure the entire network based on the topology and requirements provided. You must ensure full protocol operation, proper route advertisement, optimal path selection, and adherence to best practices.

All configurations must be verified to ensure correct routing behavior and end-to-end functionality where applicable.

This scenario reflects a real-world enterprise environment where multiple routing protocols coexist, and proper design decisions directly impact network performance, scalability, and stability.



Address Table			
Device	**Interface**	**IP Address**	**Subnet Mask**
R0	**E0/1**	**10.10.2.1**	**255.255.255.252**
	E0/2	**10.10.0.1**	**255.255.255.252**
	Loopback 0	**10.10.10.1**	**255.255.255.0**
R1	**E0/0**	**10.10.12.1**	**255.255.255.252**
	E0/1	**10.10.13.1**	**255.255.255.252**
	E0/2	**10.10.0.2**	**255.255.255.252**
	Loopback 0	**10.10.11.1**	**255.255.255.0**
R2	**E0/0**	**10.10.12.2**	**255.255.255.252**
	E0/1	**10.10.2.2**	**255.255.255.252**
	E0/2	**10.10.23.1**	**255.255.255.252**
	Loopback 0	**10.10.20.1**	**255.255.255.0**
R3	**E0/0**	**10.20.34.1**	**255.255.255.252**
	E0/1	**10.10.13.2**	**255.255.255.252**
	E0/2	**10.10.23.2**	**255.255.255.252**
	E0/3	**10.20.34.5**	**255.255.255.252**
	Loopback 0	**10.10.30.1**	**255.255.255.0**
	Loopback 1	**10.10.40.1**	**255.255.255.0**
R4	**E0/0**	**10.20.34.2**	**255.255.255.252**
	E0/1	**10.20.45.1**	**255.255.255.252**
	E0/2	**10.20.46.1**	**255.255.255.252**
	E0/3	**10.20.34.6**	**255.255.255.252**
	Loopback 0	**20.20.4.4**	**255.255.255.255**
R5	**E0/0**	**10.20.56.1**	**255.255.255.252**
	E0/1	**10.20.45.2**	**255.255.255.252**
	E0/2	**10.30.57.1**	**255.255.255.252**
	Loopback 0	**20.20.5.5**	**255.255.255.255**
R6	**E0/0**	**10.20.56.2**	**255.255.255.252**

	E0/2	10.20.46.2	255.255.255.252
	Loopback 0	20.20.6.6	255.255.255.255
R7	E0/1	10.30.79.1	255.255.255.252
	E0/2	10.30.57.2	255.255.255.252
	E0/3	10.30.78.1	255.255.255.252
R8	E0/0	10.30.89.1	255.255.255.252
	E0/1	192.168.18.1	255.255.255.0
	E0/3	10.30.78.2	255.255.255.252
	Loopback 0	192.168.8.1	255.255.255.0
R9	E0/0	10.30.89.2	255.255.255.252
	E0/1	10.30.79.2	255.255.255.252
	Loopback 0	192.168.9.1	255.255.255.0
PC	eth0	192.168.18.2	255.255.255.0

PRE-LAB CHECKLIST:

Before starting the lab, make sure everything below is done and verified

Check the IP addressing on all routers or You can copy and paste the configurations form the txt file to your routers to begin.

Task 1: Configure OSPF

All configurations must be done in **R0**, **R1**, **R2**, **R3**

1. Configure **OSPF process 10** on all required routers by placing:
   - R0,R1,R2 in **Area 10**
   - R1,R3 link in **Area 0**
2. Set the **reference bandwidth** to distinguish between **Gigabit Ethernet and Fast Ethernet interfaces**.
3. Use the **loopback 0 as router-id**.
4. Ensure R1 becomes an ABR between Area 10 and Area 0.
5. Advertise all loopback interfaces correctly in their appropriate areas.
6. Summarize Area 10 networks toward Area 0 on R1.

Task 1: Verification Commands

- ✓ show ip ospf
- ✓ show ip route
- ✓ show ip ospf database summary

Task 2: Configure eBGP

All configurations must be done in **R3**, **R4**

1. Configure eBGP between R3 (**AS 10**) and R4 (**AS 200**).
2. Configure the second eBGP session between **R3** and **AS 200**.
3. Advertise only summarized OSPF routes to AS 200.

Task 2: Verification Commands

- ✓ show ip bgp summary
- ✓ show ip bgp

Task 3: Configure iBGP Core

All configurations must be done in **R4**, **R5**, **R6**

1. configure **eigrp using AS 200** to connect the **loopback 0** between the routers.
2. Configure full-mesh iBGP between **R4**, **R5**, and **R6**.
3. Use the **loopback 0 as router-id**.
4. Use Loopback0 as update-source for all iBGP neighbors.
5. Each router must advertise **ONLY** its **Loopback1** network.

Task 3: Verification Commands

- ✓ show ip bgp
- ✓ show ip bgp neighbors

Task 4: Configure BGP Path Manipulation

Inbound traffic from AS10 must **prefer R4 e0/3**.

1. Manipulate preference of the bath between R3 and R4 to **prefer e0/3 over e0/0**.
   - Use Prefix-list named **PREFERRD_LINK.**
   - using Route map name **PREFERRD_PATH.**

Task 4: Verification Commands

- ✓ show ip bgp
- ✓ show ip route
- ✓ traceroute tests

Task 5: Configure Advanced EIGRP

1. Configure Named **EIGRP Branch1 with AS 300** on **R7**, **R8**, and **R9**.
2. Advertise **all networks and loopback**.
3. Configure authentication for eigrp links using key chain
4. Create key chain with the **name <Router name>-EIGRP use EIGRP@KEY for key string and sha 256.**
5. Configure R8 as an EIGRP stub (**connected + summary**).

Task 5: Verification Commands

- ✓ show ip eigrp neighbors detail
- ✓ show ip route
- ✓ show ip eigrp topology

GOOD LUCK :)